4.7.51

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Problem 1 (4.7.51)**.** *Find the equation of the plane passing through the point (-1, 3, 2) and perpendicular to each of the planes* $x + 2y + 3z = 5$ *and* $3x + 3y + z = 0$.

Solution: The normal vectors to the given planes are $\mathbf{n_1} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ and $\mathbf{n_2} = \begin{pmatrix} 3 \\ 3 \\ 1 \end{pmatrix}$.

Let the normal vector of the required plane be $\mathbf{n} = \begin{pmatrix} n_1 \\ n_2 \\ n_3 \end{pmatrix}$. Since the required plane is perpendicular to both given planes, its normal vector $\mathbf{n}$ must be orthogonal to both $\mathbf{n_1}$ and $\mathbf{n_2}$. This gives the following homogeneous system of linear equations:

$$\mathbf{n_1}^\top \mathbf{n} = n_1 + 2n_2 + 3n_3 = 0 \tag{0.1}$$

$$\mathbf{n_2}^\top \mathbf{n} = 3n_1 + 3n_2 + n_3 = 0 \tag{0.2}$$

To solve this system, we form the augmented matrix and reduce it. The final reduced row echelon form is:

$$\left(\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 3 & 3 & 1 & 0 \end{array}\right) \xrightarrow{\text{Row Reduce}} \left(\begin{array}{ccc|c} 1 & 0 & -7/3 & 0 \\ 0 & 1 & 8/3 & 0 \end{array}\right) \tag{0.3}$$

From the reduced form, we have:

$$n_1 - \frac{7}{3}n_3 = 0 \implies n_1 = \frac{7}{3}n_3$$

$$n_2 + \frac{8}{3}n_3 = 0 \implies n_2 = -\frac{8}{3}n_3$$

The solution space for $\mathbf{n}$ is spanned by the vector $\begin{pmatrix} 7/3 \\ -8/3 \\ 1 \end{pmatrix}$. We can choose any non-zero scalar multiple of this vector for our normal. Letting the free variable $n_3 = -3$ gives a simple integer vector:

$$\mathbf{n} = \begin{pmatrix} -7 \\ 8 \\ -3 \end{pmatrix} \tag{0.4}$$

The equation of the required plane passes through the point $\mathbf{P_0} = \begin{pmatrix} -1 \\ 3 \\ 2 \end{pmatrix}$ and has the normal vector $\mathbf{n}$. The equation is $\mathbf{n}^\top \mathbf{p} = \mathbf{n}^\top \mathbf{P_0}$.

$$c = \begin{pmatrix} -7 & 8 & -3 \end{pmatrix}^\top \begin{pmatrix} -1 \\ 3 \\ 2 \end{pmatrix} = 7 + 24 - 6 = 25 \tag{0.5}$$

Thus, the equation of the plane is:

$$\begin{pmatrix} 7 & -8 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = -25 \tag{0.6}$$

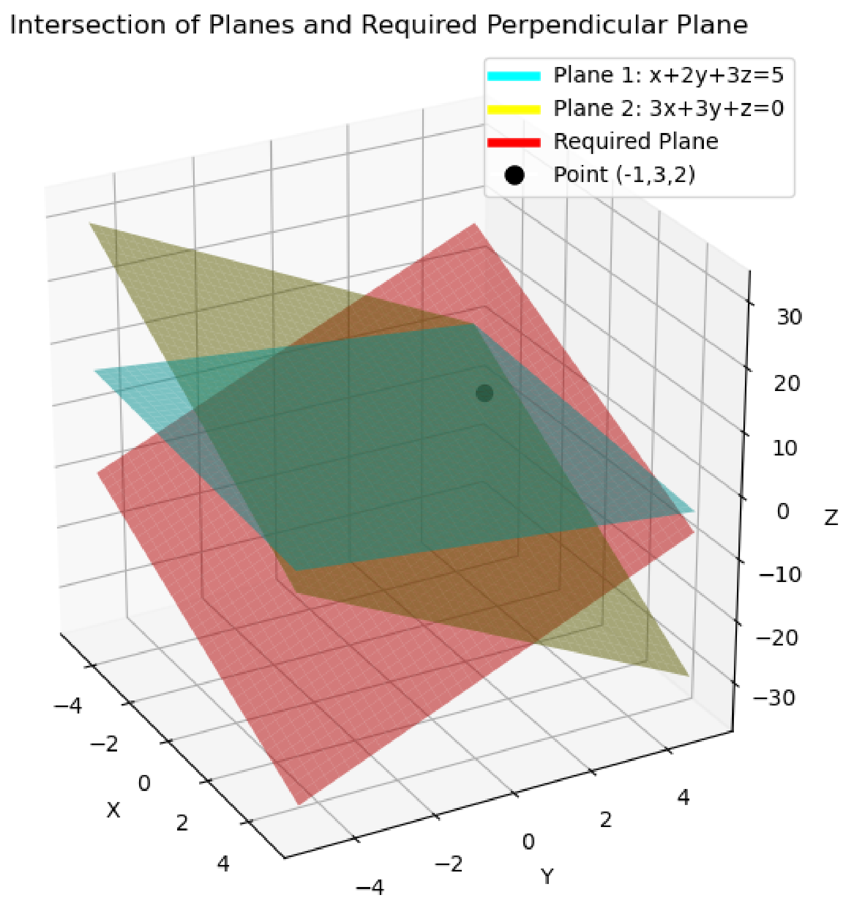
Intersection of Planes and Required Perpendicular Plane
Plane 1: x+2y+3z=5
Plane 2: 3x+3y+z=0
Required Plane
Point (-1,3,2)
X
Y
Z
−4
−2
0
2
4
30
20
10
0
−10
−20
−30

Fig. 0.1